

Atomistic origins of anharmonic lattice dynamics and thermal expansion in perovskite photovoltaics

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 Check for updatesJulian A. Steele ^{1,2} 

Metal halide perovskites are swiftly becoming a leading class of photovoltaic materials, yet unlike conventional semiconductors, they exhibit highly anharmonic lattice vibrations that produce extreme rates of thermal expansion and substantial mismatch with other device layers. These effects are exacerbated under natural day/night cycling, where repeated heating and cooling drive cumulative stress, defect generation and accelerated degradation. Here we connect the atomistic origins of anharmonic lattice dynamics in perovskites with their macroscopic thermo-mechanical properties, from phonon–phonon interactions and local disorder to complex thermal–phase relations. We assess how both the degree of anharmonicity and thermal expansion rates vary across temperature, composition and phase, and highlight the emergence of anisotropic and even negative thermal expansion in low-symmetry perovskite phases. By linking fundamental physics to device-level challenges, we provide a framework for engineering durable perovskite absorbers and outline promising approaches to regulate thermal strain in the pursuit of long-lived, commercially viable solar cell technologies.

The rise of metal halide perovskites (MHPs) has distinctly disrupted contemporary photovoltaic research^{1,2}, as they combine simple synthetic methods with high-quality optoelectronic performance and promising photo-conversion efficiencies (PCEs). Moreover, the materials are abundant, and their ease of production implies low-cost and high-throughput manufacturing. Whereas the properties of MHPs resemble traditional semiconductors, such as III–V materials and Si, their soft and dynamic crystal structure confers an assortment of complex thermally activated properties³—from ion (or defect) transport^{4,5} and polaron formation to local polar fluctuations⁶ and complex thermal–phase relations⁷.

Underpinning these interesting properties is the tendency of perovskites to exhibit fluctuations in their local crystal structure⁶ and strong lattice anharmonicity⁸—deviations from the harmonic approximation for oscillating atoms. This is because there exists both a ‘flexible’ network of corner-sharing metal halide octahedra and an

ionic sublattice of caged cations, with varying degrees of freedom and interactions. At elevated temperatures, a static picture of the perovskite solar cell (PSC) structure simply breaks down and one must account for anharmonic lattice dynamics to understand its bulk physical properties.

This notion extends to the mechanical properties of MHPs, which ideally should be tailored for specific applications and engineered to withstand thermal-induced stress, given that thermal expansion in solids is driven by lattice anharmonicity. For example, in contrast to Si, the rates of thermal expansion (thermal expansivity) found in anharmonic MHP crystals are far larger than many of the other layers in a device, straining the absorber during each day/night thermal cycle (Fig. 1a), and posing a unique and serious problem for PSCs^{9,10}. Given that natural day/night cycling involves changes in both illumination and temperature, Shen et al.¹¹ decoupled their respective impacts and found that cycled illumination with a fixed temperature does not inevitably

¹Australian Institute for Bioengineering and Nanotechnology (AIBN), The University of Queensland, Brisbane, Queensland, Australia. ²School of Mathematics and Physics, The University of Queensland, Brisbane, Queensland, Australia. ✉ e-mail: julian.steele@uq.edu.au

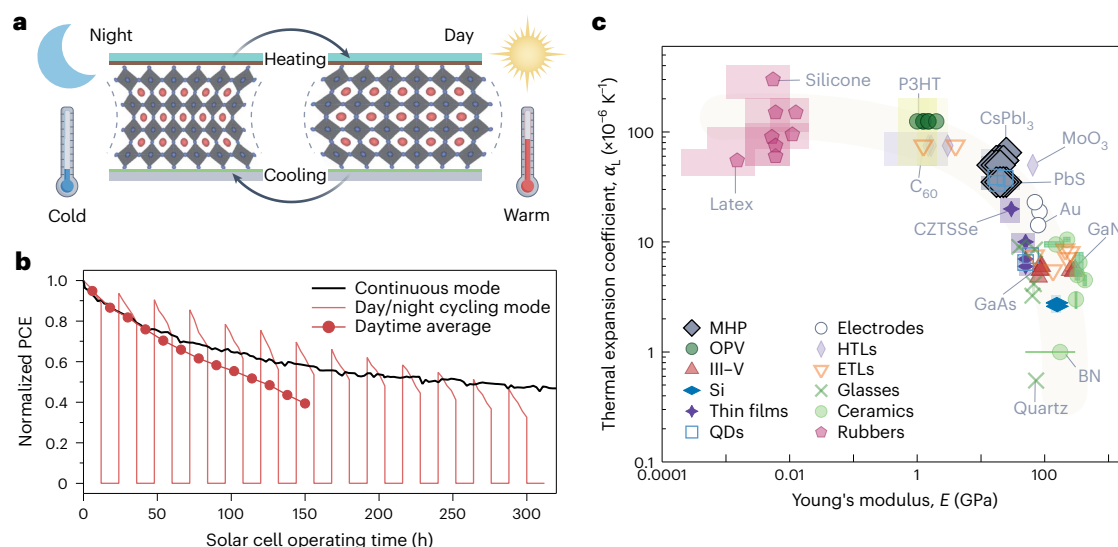


Fig. 1 | Thermal cycling of the perovskite absorber. **a**, Illustration of the perovskite structure, which periodically contracts during the night and expands during day, with each day/night thermal cycle. **b**, Stability of FAPbI₃-based solar cells (FA = formamidinium) working under continuous illumination (55 °C) and day/night cycling modes (room temperature at night and 55 °C during the day, while the illumination is kept constant). The red circles represent the average value for the cycling mode, with the operating time being accumulated only during the daytime portion of the cycle. Data from ref. 11. **c**, Linear thermal expansion ($\alpha_L = \frac{1}{L} \frac{dL}{dT}$, where L is length and T is temperature) of different materials as a function of their Young's modulus (E)^{47,51,81,85,96,106–138}. For a material,

$E = \sigma/\epsilon$, which represents the ratio of the tensile stress force σ (in units of Pa) to the strain ϵ (that is, the relative change in length in the material's elastic response region: $\epsilon = \Delta L/L_0$, a dimensionless parameter, where L_0 denotes the initial length). Common hole transport layer (HTL) and electron transport layer (ETL) materials are included, along with metallic electrodes (Al, Ag, Au), alongside MHPs. Some representative materials are labelled, and the shaded boxes represent the value spread across available studies. The thick, light-blue background curve simply acts as an aid for the eye to resolve the general trend between these two physical properties. OPV, organic photovoltaic; QDs, quantum dots; P3HT, poly (3-hexylthiophene); CZTSSe, Cu₂ZnSn(S,Se)₄. Panel a created with VESTA¹³⁹.

lead to pronounced accelerated degradation¹². Considering the role of thermo-mechanical stresses alone¹³, the introduction of excessive stress, that is, beyond the recoverable range of the perovskite crystal¹⁴, does accumulate cracks and interfacial defects in the perovskite layer over time, accelerating performance decay¹¹ (Fig. 1b). Notably, the main sources of pronounced PCE loss under repeated thermal cycling have been assigned to steady reductions in the fill factor (FF) and open-circuit voltage (V_{oc})⁹. This indicates that the widely adopted¹⁵ accelerated testing protocols for traditional solar cells—that typically involve exposing devices to sustained light and/or heat—are simply inadequate for materials such as perovskites.

Here we aim to connect the complex, microscopic world of perovskite crystals, that is, anharmonic lattice vibrations, with their large rates of macroscopic thermal expansion. Given that thermal strain has been identified as a key stumbling block towards realizing long-term operating lifetimes^{9,11}, understanding the origins of thermal expansion in MHPs is particularly meaningful for the development of commercial devices. In this spirit, we finish with a consolidated overview of effective strategies to curb the effects of thermal strain in PSCs and propose testing protocols to help explain this intrinsic degradation pathway.

Thermo-mechanical properties of the absorber

Young's modulus (E), which is a measure of material stiffness, and the coefficient of linear thermal expansion (α_L) are perhaps the most fundamental of properties for analysing the durability of a PSC under thermal cycling. Figure 1c shows these two properties across various material categories, including comparisons with different solar cell technologies, their components and the extremes of (stiff) glasses and (flexible) rubbers. Whereas a slight variation exists across different perovskite compositions, what is striking is their relative position to other materials at the inflection point of the trend; MHPs exhibit both a relatively high Young's modulus and a high rate of thermal expansion. In other words, they expand/contract readily under temperature differentials but have a poor capacity to accommodate stress before fracturing.

Managing thermal stresses means managing the large mismatch in the rates of thermal expansion between the different layers of the device^{9,16–18}. With the layers clamped together, the perovskite film is deformed (via expansion/contraction) in directions perpendicular to the strain loading, and is described by Poisson's ratio: $\nu = -d\epsilon_x/d\epsilon_y$, where ϵ_x and ϵ_y represent the strain in orthogonal directions. The relationship between stress (σ) and a change in temperature (ΔT) is described by:

$$\sigma(\Delta T) = \frac{E_p}{1 - \nu_p} (\alpha_s - \alpha_p) \Delta T, \quad (1)$$

where the subscripts P and S represent the perovskite and substrate materials, respectively, and $(\alpha_s - \alpha_p)$ denotes the mismatch in thermal expansion between the two.

To realize durable PSCs, equation (1) implies that thermo-mechanical regulation is needed to reduce the misfit strain. For instance, decreasing lattice anharmonicity will typically reduce thermal expansion rates and increase the Young's modulus.

Anharmonic lattice dynamics in perovskite crystals

We can in fact connect the simple diatomic model presented in the figure of Box 1 to Fig. 1c; the Young's modulus is related to the curvature of the trough, whereas the thermal expansion is related to the asymmetry of the potential energy surface (PES). Put differently, stronger bonds have steeper and narrower potential wells, resulting in smaller thermal expansion rates. This underscores the fact that the thermal durability of perovskite absorbers is fundamentally limited by our ability to understand and control lattice anharmonicity.

Anharmonic structural instabilities and defects

There are many different types of dynamic distortion that can perturb the ideal perovskite structure (Fig. 2a). For example, Worhatch et al.¹⁹ used X-ray pair distribution function analysis to conclude that the

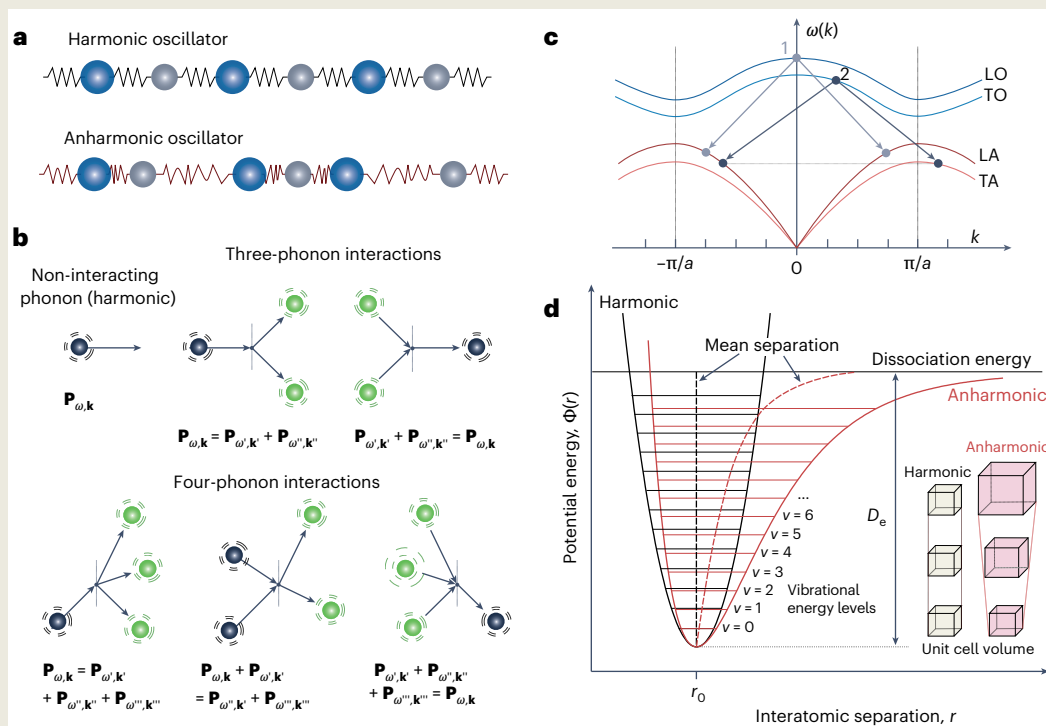
BOX 1

Strong anharmonicity drives large rates of thermal expansion in perovskites

The figure illustrates the underlying origins of thermal expansion in perovskites—anharmonic lattice vibrations. In the harmonic approximation of a simple one-dimensional (1D) chain of vibrating atoms (a), they are bound by linear forces, and phonon modes are established between neighbouring atoms. In the real-world, however, the interatomic forces are non-linear, and different primary types of bond within perovskite crystals exhibit distinct characteristics, for example, ionic, covalent, metallic, van der Waals¹⁴³ and hydrogen bonding. In general, stiffer bonds present weaker anharmonic effects, and hierarchical bonding increases the anharmonicity substantially. Through anharmonic interactions, a single phonon mode induces a periodic elastic strain that modulates the elastic constants between atoms in space and in time (a). A second phonon interacts with the modulation and is scattered via multi-phonon interactions (b). Over time, multi-phonon processes induce phonon decay, and the energy

and momentum are redistributed within the lattice (c). Compared with traditional semiconductors, strong phonon–phonon interactions in MHPs endow them with several interesting thermal properties⁸², such as low thermal conductivity and low heat capacity.

Anharmonic interactions imply asymmetry in the space domain of vibrating atoms, pushing neighbouring atoms further apart and inducing thermal expansion. To illustrate the point, we consider a simple diatomic model (d), where the vibrations supported by the harmonic potential are non-interacting and the average atomic position is unchanging with temperature, that is, zero thermal expansion. The anharmonic terms describing higher-order phonon–phonon interactions lead to asymmetry in the PES (d). When higher vibrational states are populated through rising temperatures, larger atomic displacement amplitudes force atoms to spend more time further apart, causing thermal expansion.



Box fig. 1 a, Illustration of a 1D chain model comprising vibrating atomic masses bound by linear (harmonic) and non-linear (anharmonic) bonding potentials. **b**, Vectorial representation of different types of multi-phonon process and momentum exchange ($\mathbf{P}$) between interacting phonons with frequencies, ω , and wave vectors, $\mathbf{k}$. **c**, Examples of anharmonic phonon decay processes within the representative phonon band diagram of a polar crystal. (1) The longitudinal optical (LO) phonon decays into two longitudinal acoustic (LA) phonons; (2) the transverse optical (TO) phonon decays into LA and transverse acoustic (TA) phonons at slightly different wave vector (k) values. The horizontal dotted line represents the

Umklapp process, where the wave vector of the TA phonon is created outside the first Brillouin zone (that is, from $-\pi/a$ to π/a). ω , phonon frequency. **d**, The potential energy of the atoms, Φ , as a function of their interatomic displacement, r , calculated using the Morse oscillator model¹⁴⁴. Near the minimum (around the equilibrium interatomic separation r_0), the harmonic potential is quite similar to the anharmonic one. Elevated energy vibrational levels are identified as $v=1, 2, 3$ and so on. For the anharmonic case, increases in the vibrational level translate into increases in the unit cell volume, as illustrated in the inset.

local cubic symmetry in MASnBr_3 (MA = methylammonium) is in fact broken via dynamic octahedral tilting. Such distortions are linked to anharmonic phonon-induced polar fluctuations of the metal halide

sublattice⁶, and are governed by the size-dependent lattice geometry at a certain temperature, rather than by the chemical nature of the A-site cation. Whereas it has been proposed that the strong anharmonic tilting

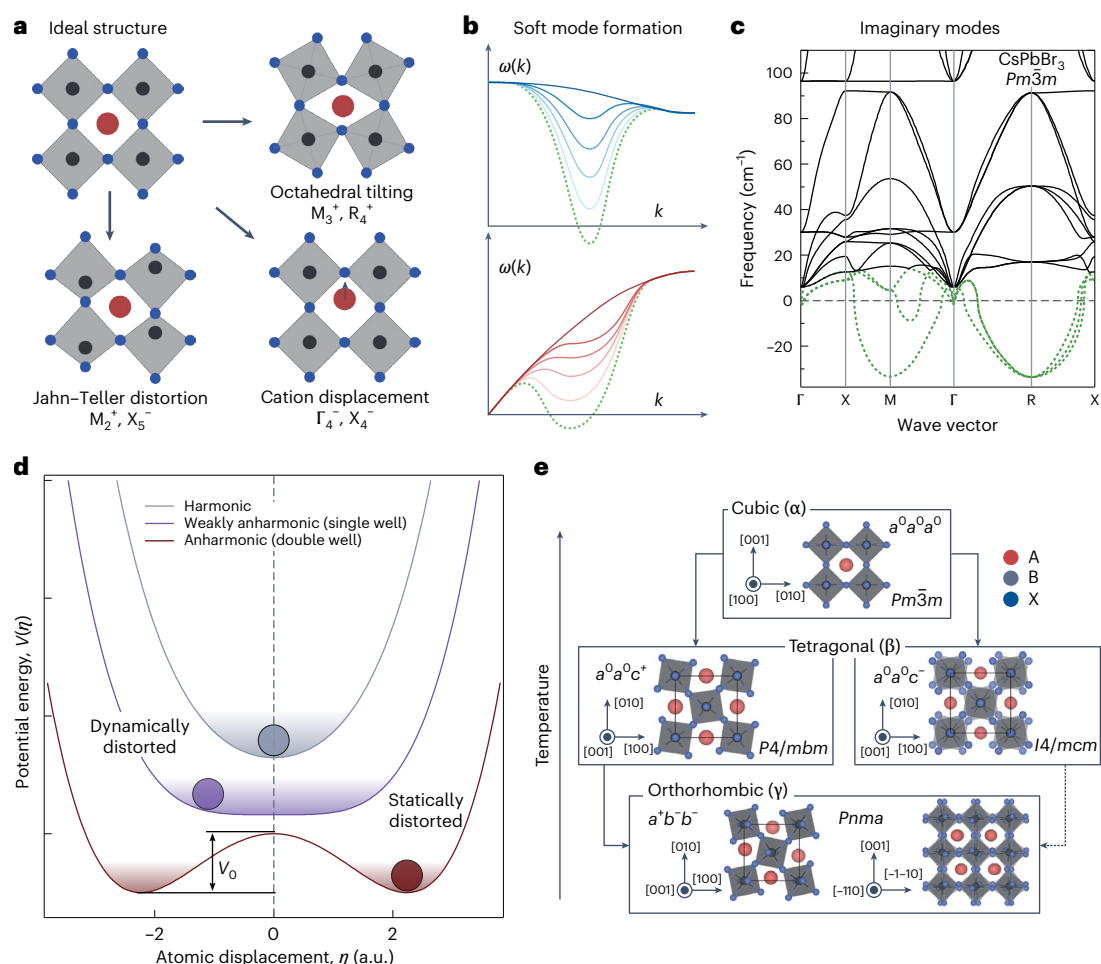


Fig. 2 | Anharmonic instabilities and phase transitions. **a**, Illustrative visualization of the three different general classes of distortion exhibited by the perovskite structure and their corresponding irreducible representations from group theory. Here the A-site cations are red, the metal atoms are dark grey and the halide ions are blue. **b**, Illustration of phonon softening with changes in temperature for optical (top) and acoustic (bottom) vibrations. The dashed lines represent when the mode frequencies approach zero and become dispersive in nature, triggering a phase transition. **c**, The calculated phonon band structure of cubic CsPbBr₃ perovskite using the harmonic approximation. The imaginary phonon modes (occupying negative frequencies) are shown as dashed lines.

Data from ref. 75. **d**, Depiction of the PES for various levels of anharmonicity, represented by a generalized¹⁴⁰ quartic anharmonic-oscillator potential energy of the form $V(\eta) = a\eta^2 + b\eta^4$, where η is the displacement parameter, $b > 0$ and a is the level of anharmonicity. In these examples, $a > 0$ (grey), $a \approx 0$ (purple) and $a < 0$ (brown). **e**, Commonly observed MHP crystal structures following group-subgroup symmetry relations. Glazer's notation¹⁴¹ is used for the tilts, which classifies octahedral distortions that translate through the crystal during thermal expansion. Tilt identification can be understood using the subsequent octahedral layers being stacked along the tilt axis in-phase (+) or antiphase (-). Panels adapted with permission from: **a**, ref. 22, AIP; **e**, ref. 142, AIP.

exhibited by Pb(II)- and Sn(II)-based perovskites is largely derived from their lone pair orbital configurations²⁰ and related Jahn-Teller-like effects (Fig. 2a), Caicedo-Dávila et al.²¹ recently showed through first-principles calculations that such cations are not a prerequisite and identified the main determining factor to be the flexible nature of the octahedral network.

All types of structural distortion²² shown in Fig. 2a are vector quantities, modifying the anharmonicity in certain crystallographic directions and establishing anisotropic properties. Moreover, they can couple via phonon-phonon interactions. For instance, both MA- and FA-containing perovskites are anticipated to show strong coupling between the orientations of the molecular cations and the local displacements of the inorganic sublattice²³, indicating a mixed order-disorder/displacive nature for transitions between different perovskite phases. Yet, little is known of the general impact of lattice defects on the rates of thermal expansion; the modified spring constants that are around defects change the anharmonicity and normal modes, and result in changes to the macroscopic thermal expansion rates when in high concentrations²⁴. For example, Dobrovolsky

et al.²⁵ linked excess defects in MAPbI₃ single crystals to increases in the tetragonal-to-orthorhombic phase-transition temperature. As the absorber naturally ages and accumulates defects, this concept becomes important for realizing effective device engineering principles.

Soft modes and imaginary phonons

Phonon-phonon interactions modify the MHP vibrational spectrum and can lead to a large frequency reduction for a particular mode (so-called 'mode softening') near phase transitions or critical lattice instabilities (Fig. 2b). When the atomic displacements become increasingly asynchronous, the restoring forces that stabilize the structure weaken and the energy required to excite the anharmonic phonons becomes very low, such that the lattice distorts along the soft mode vector. In PSCs, the atomic displacements can coordinate and generate collective long-range displacement patterns and dictate new equilibrium positions for the atoms, breaking the previous crystal symmetry. The interaction between soft modes and other phonons will affect the thermo-mechanical properties of the perovskite, and its

corresponding Young's modulus will decrease as the lattice becomes more compliant to the deformations associated with the lattice vibration. Notably, elastic behaviour, such as the relationship between stress and strain, can become highly non-linear due to the large atomic displacements. For example, Ge et al.²⁶ showed that MAPbI₃ shows an order-of-magnitude increase in the rate of linear thermal expansion (along <001>) near its tetragonal-to-cubic phase-transition temperature at -55°C . This becomes highly problematic if such instabilities occur inside the operating temperature range¹⁵ of a device, as large non-linear thermo-mechanical perturbations are introduced through each thermal cycle.

The theoretical frameworks used to describe nuclear dynamics in perovskites are often based on the harmonic approximation²⁷. However, such calculations become inaccurate when simulating experimentally determined perovskite structures. Soft modes often emerge in the ab initio calculations of the harmonic phonon dispersion²⁸ of anharmonic perovskite structures^{27,29}, whereby the calculated vibrational spectrum yields imaginary phonons with negative frequencies (Fig. 2c). This typically indicates structural instability; if the atoms follow this vibrational pattern, the crystal will become distorted and the displacements will reduce the potential energy of the system. This is in contrast to dynamically (mechanically) stable crystals, which always increase their potential energy with any combination of atomic displacements.

There are ways to refine the first-principle calculation to mitigate imaginary modes^{29,30}, yet their appearance in harmonic models of anharmonic structures can yield important insights. For the example of the calculated phonon dispersion of cubic CsPbBr₃ in Fig. 2c, freezing the imaginary tilting mode along the M point will transition the crystal into its lower-symmetry perovskite phase (Fig. 2a). Consequently, the higher-symmetry cubic and tetragonal phases of halide perovskites are predicted to be dynamically unstable at elevated temperatures^{31,32} and are only resolved through time-averaged studies of the atomic structure, for example, those derived from neutron diffraction^{33–35} and X-ray diffraction³⁶.

Local disorder

The fundamental mismatch between an experimentally determined structure and its true local structure highlights the importance of first-principles materials modelling techniques to account for anharmonicity and attain accurate atomic-scale models. Within this context, tremendous progress has been made using computational tools²⁹ to interpret anharmonic phonon modes, temperature dependence and lattice expansion in MHPs. The self-consistent phonon (SCP) theory, introduced by Hooton³⁷, provides an effective strategy for calculating anharmonic effects in crystals by replacing the complex, anharmonic potential with an effective, temperature-dependent harmonic potential that minimizes the free energy of the system. Furthermore, using the SCP theory via special displacements³⁸ opens the way for understanding the vibrational and temperature-dependent properties of different perovskite compositions. Patrick et al.³⁹ explained the unstable nature of cubic α -phase of CsSnI₃ within the harmonic approximation is due to rotational instabilities of the SnI₆ octahedra. Subsequently, using the SCP theory, these atomic displacements form soft modes that are stabilized via anharmonic phonon–phonon coupling between the Cs ions and their iodine cages. Thus, the emerging physical picture is that the BX₆ octahedral sublattice has an average structure with high symmetry, while the local structural motifs have low symmetries. In reality, this can result in unique grain micro-structure, as dynamic cooperative rotations of PbX₆ octahedra introduce twin-like boundaries and establish subdomains—even within a ‘single crystal’⁴⁰. At the device level, the formation of small strained subdomains that expand in different directions, at different rates, is a likely source of compromised interfacial binding and delamination of perovskite grains^{9,41}.

Simulations of dynamic instabilities in the octahedral network lead to the non-thermal energy minimization of perovskite structures

with a so-called ‘polymorphous’ nature⁴², characterized by correlated local disorder that is not captured experimentally using standard diffraction methods. By comparing the simulated physical properties, for example, electronic bandgap, phonon dispersion, carrier transport and so on, of polymorphous (locally disordered) and monomorphous (ideal high-symmetry cubic) perovskite structures, a number of recent studies^{38,42–44} have pointed out that accounting for anharmonicity and polymorphism is vital for generating theoretical findings that agree with experiment.

This concept also extends to thermal expansion behaviour, with the quasiharmonic approximation⁴⁵ finding great utility in describing volume-dependent thermal effects in MHPs. For instance, it has been used to determine that composition-dependent anharmonicity emerges as a central factor when simulating expansion rates in MHPs of the same phase⁴⁶. A vivid example of this relates to the role of *ns*² lone pair orbitals in Sn²⁺ and Pb²⁺ compositions, which promote stereochemically off-centre atomic motions²⁰. Consequently, the chemically more active tin lone pairs probably account for experimentally determined thermal expansion rates that are relatively larger than their Pb-based counterparts⁴⁷.

Atomic displacements and phase transitions

Thermal excitations lead to the occupation of higher vibrational energy levels and the atomic probability density is smeared over the PES (Fig. 2d). Promising developments have been made in recent years towards developing first-principles models of the PES^{29,38,48,49} to analyse their relative shapes and depths. Whereas the PES examples provided in Fig. 2d oversimplify the problem for different MHP compositions²², they do well to explain their similarly rich phase space⁵⁰. When anharmonic distortions transition from dynamic and static ones, new atomic displacements can coordinate collective long-range patterns, breaking the previous (average) crystal symmetry. In general, MHPs undertake transitions to lower-symmetry phases at reduced temperatures⁷; Fig. 2e tracks a perovskite crystal^{51,52} through its high-temperature cubic α -phase, which transitions upon cooling to a tetragonal β -phase and an orthorhombic γ -phase. Owing to recent advances in molecular dynamics simulations, and the use of machine-learned force fields based on first-principles data, it is now possible to accurately reproduce the thermal–phase domains of different MHP compositions^{53–56}.

A-site dynamics

Even within a single perovskite phase, the local crystal structure fluctuates through so-called ‘dynamic disorder’^{31,57,58} of the caged cations. Early studies of dynamic disorder used complex dielectric measurements⁵⁹, nuclear magnetic resonance spectroscopy⁶⁰ and millimetre-wave spectroscopy⁵⁷, to determine that the caged organic cations undergo tumbling (Fig. 3a) between various dipole orientations when thermally activated⁵⁹. At reduced temperature, the cations are further locked in specific orientations due to strong hydrogen bonding with the halide ions and have been described via a crystal–liquid duality⁶¹. It follows that the A-site dynamics in hybrid organic–inorganic perovskites are critical to understanding the changes in thermo-mechanical properties when fashioned into devices. With MA-based perovskites representing an active area of study historically⁶², although becoming less popular recently due to intrinsic stability issues⁶³, we consider them here as a useful case study to explore the role of A-site dynamics.

Figure 3b compares the temperature-dependent dielectric constant (ϵ') measurements carried out by Govinda et al.^{64,65} on disparate compositions to identify the contributions of caged A-site cations. There exist clear qualitative differences in the thermal behaviours of the FA⁺ and MA⁺ organic moieties, as strong temperature dependence is only expressed for the MA-based perovskite due to changes in its rotational degrees of freedom across the orthorhombic-to-tetragonal phase transition⁶⁶. Conversely, the absence of any strong temperature dependence in FAPbBr₃ indicates that the FA dipoles are in a frozen

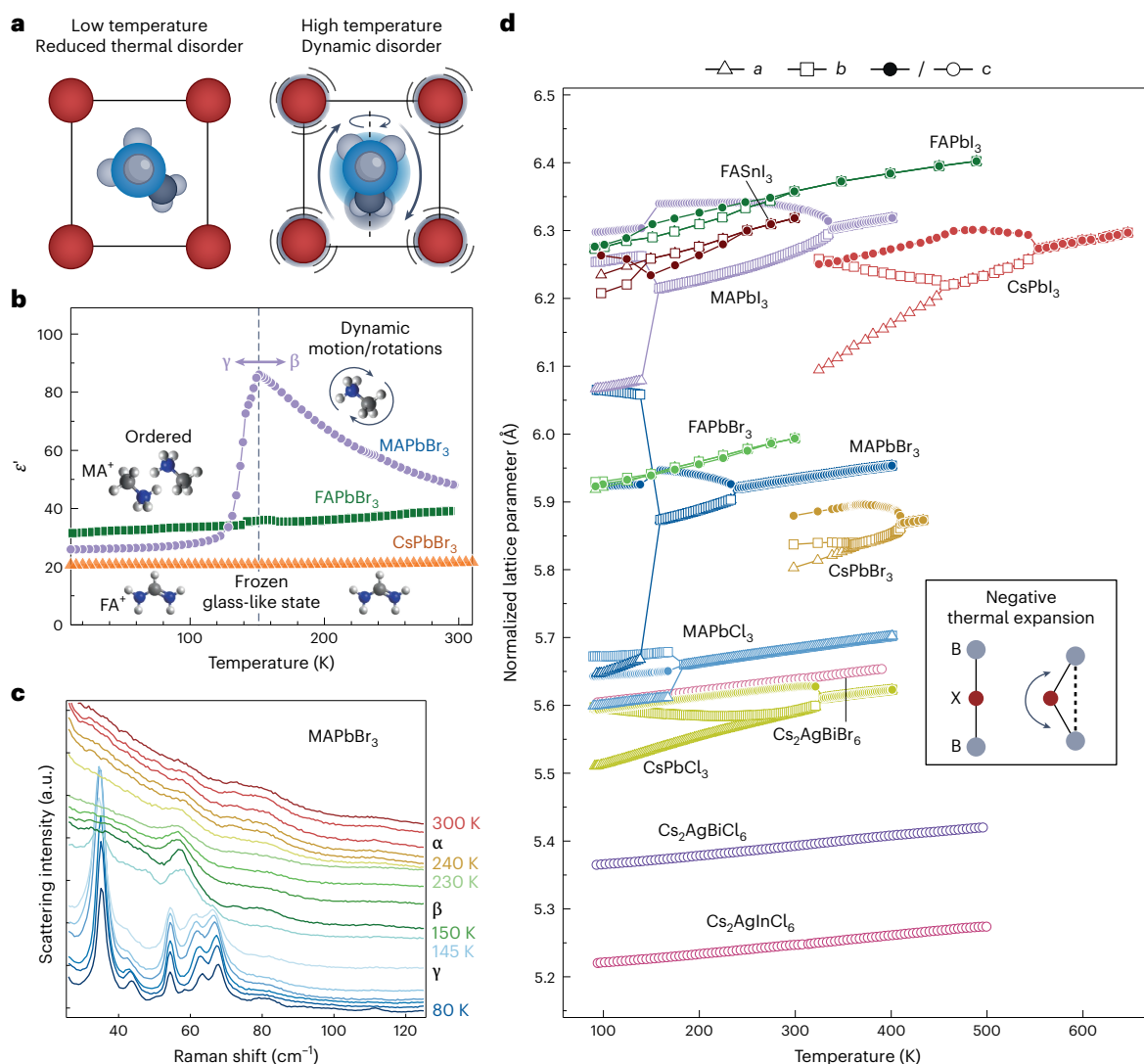


Fig. 3 | Temperature-dependent disorder and lattice parameters. **a**, Illustration of thermally activated dynamic disorder of the caged organic cation (MA⁺; blue and grey) inside the inorganic halide cage (red) of the perovskite structure. **b**, Comparison of the temperature-dependent dielectric constant (ϵ') measured for the MAPbBr₃, FAPbBr₃ and CsPbBr₃ perovskites with an AC frequency of 1.08 MHz. The insets show illustrations of the general (dis)order expressed by the different organic cations. Data from refs. 64,65. **c**, Temperature-dependent Raman spectral evolution of an MAPbBr₃ single crystal through its different phases, focusing on the low-energy modes that arise mainly from the metal halide sublattice. Data from ref. 73. **d**, Comparison of the

temperature-dependent pseudo-cubic lattice parameters (a , b and c) of different MHP compositions^{47,51,81,85,123}. Both open and filled circles have been used for the c axis to enable better contrast between overlapping data points. Each of the pseudo-cubic lattice parameters have been normalized in relation to the archetypal parent unit cell (cubic) through rescaling factors of $\alpha(a, a, a)$, $\beta(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}, c)$ and $\gamma(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}}, \frac{c}{2})$. The inset shows a representation of the local vibrational modes in an ABX₃ perovskite (involving B–X–B bonds) that give rise to NTE behaviour. These modes lead to a decrease in the observed metal–metal distance (dashed line) in a given direction. Inset in **d** adapted with permission from ref. 80, APS.

glass-like state. Regarding the reported²⁶ sharp rise in thermal expansion in MAPbI₃ perovskite at its tetragonal-to-cubic phase transition, coupling between the caged A-site organic cations and the inorganic backbone phonons cause such anomalous thermo-mechanical behaviour²⁶.

Probing phonon anharmonicity

Changes to the phonon spectrum are a microscopic reflection of anharmonic effects, which lead to macroscopic changes in the elastic moduli and thermal expansion. Direct measurements of the vibrational spectrum are the most powerful and straightforward, that is, via techniques such as infrared spectroscopy⁶⁷, Raman spectroscopy⁶⁸ and inelastic electron⁶⁶ and neutron⁶⁹ scattering. Complementing the dielectric measurements in Fig. 3b, the unique thermal behaviour in MAPbBr₃ is vividly captured in the temperature-dependent Raman scattering data

shown in Fig. 3c. There exist abrupt changes to the phonon spectra near the phase-transition points, indicating strong anharmonic effects⁷⁰:

- Phonon frequencies tend to soften with temperature as phonon–phonon interactions increase⁷¹.
- Phonon linewidths (Γ) correspond to the inverse phonon lifetime ($\tau \propto 1/\Gamma$)⁷², and influenced by the crystal quality and anharmonic effects, that is, high temperatures increase phonon interactions and accelerate phonon decay, broadening the Raman modes.
- Raman selection rules govern which modes are Raman-active and observed, based on the crystal symmetry. The lower-symmetry γ - and β -MAPbBr₃ perovskites yield a rich Raman-active spectrum^{73,74} (Fig. 3c), becoming less active in the higher-symmetry phases. Notably, the metal halide phonons of α -phase cubic perovskites (point symmetry: O_h) are theoretically all Raman-inactive.

Their appearance in the experimental Raman spectra is thus a signature of anharmonic distortions and local disorder which break local symmetry, relaxing the selection rules. This Raman activation mechanism was recently corroborated via *ab initio* molecular dynamics simulations of cubic CsPbBr₃ (ref. 75).

Being vector quantities, anharmonic phonons become highly anisotropic in the low-symmetry perovskite phases⁷⁶, and even more pronounced in two-dimensional layered systems^{77,78}. This is important for polycrystalline thin films containing highly oriented grains, as the thermal expansion rates of the crystal will also be dramatically different along certain crystallographic directions and must be accounted for when designing thermally robust PSCs.

Thermal expansion properties of metal halide perovskites

Anisotropic thermal expansion of the perovskite lattice

Tuning the optical bandgap of the PSC absorber represents a central engineering parameter, albeit through X-site mixing for wide-bandgap top cells in tandem designs, narrow-bandgap materials using a variety of A- or B-site mixing for a single-junction absorber or within all-inorganic Pb-free double-perovskite devices. From a conceptual standpoint, pure ABX₃ compositions provide clear representations of how different perovskite compositions and structures evolve across their thermal-phase domains. Pure ABX₃ compositions also represent the majority of structural data available in the literature today, thus highlighting an important knowledge gap.

X-ray diffraction remains the most powerful technique for resolving the perovskite crystal structure³⁶, with Fig. 3d bringing together the temperature dependence of different perovskite lattice parameters. Common among ABX₃ compositions is the formation of an anisotropic phase at lower temperatures and a high-symmetry cubic phase at higher temperatures. Markedly, the unique cation dynamics of MA-containing compositions are manifested through abrupt shifts in the lattice parameters and the rates of anisotropic linear thermal expansion. Furthermore, the axes of several orthorhombic and tetragonal phases display non-linear dependences with temperature, so that the thermal expansivity becomes highly temperature-dependent.

Positive volume thermal expansion is common in all MHP compositions (Fig. 4), a property maintained across phase transitions and linked to extended thermal-driven atomic displacements (panel d in Box 1). As such, the linear negative thermal expansion (NTE) in Fig. 3d shown by the lower-symmetry phases of ABX₃ perovskites needs clarification. In short, there are no known examples of NTE in solids in which the interatomic bond lengths shorten with heating⁷⁹, and its appearance here can be viewed as a fingerprint of the anharmonic dynamical tilting. Considering the network of metal–halide–metal (B–X–B) bonds forming the octahedra in perovskites, the mechanism driving NTE involves the vibrational displacement of the metal–metal axes⁸⁰. Mu et al.⁴⁶ simulated the NTE-related vibrations in CsPbI₃ using the quasiharmonic approximation, and found that they arise from I displacements, for example, octahedral rotations and tilting. Rising temperature increases the amplitude of the displacements and results in a net contraction along particular crystallographic directions (inset of Fig. 3d). For example, the contraction of one lattice axis in the orthorhombic phase with increasing temperature is a free-energy-minimizing response, driven by anisotropic phonon behaviour and the structural tendency towards a more symmetric, higher-entropy tetragonal phase.

Determining the rates of volumetric thermal expansion

A single value, or an upper and lower limit depending on the temperature range, is often prescribed as the coefficient of volumetric thermal expansion (α_v) in MHPs. It is also sometimes calculated over temperature ranges spanning multiple perovskite phases. However, this simply lacks precision given what we have discussed so far, as a

variety of features in the thermal expansion rate emerge due to shifts in the underlying anharmonicity. The best determination of this parameter in PSCs is to use a high-order polynomial to determine the temperature-dependent rate of expansion across a single phase⁸¹. Figure 4 presents an analysis (Methods) of the temperature-dependent unit cell volumes and thermal expansion rates for a variety of different perovskite compositions.

Chemical and structural trends in thermal expansion

Few studies have explored the underlying mechanisms and chemical trends of thermal expansion rates in different PSC compositions^{46,81}. An experimental survey completed by Haeger et al.⁸² demonstrated that X-site substitution strongly influences the thermal expansion of inorganic CsPbX₃ perovskites, for example, moving from I to Br generally increases the Pb–X bonding strength and reduces anharmonicity and rates of thermal expansion^{81,83}. Sun et al.⁸⁴ determined that the Young's modulus data follow a similar trend, that is, $E_{\text{Cl}} > E_{\text{Br}} > E_{\text{I}}$. Conversely, when the A-site species is replaced with an organic cation with larger ionic radius, the thermal expansion of halide perovskites becomes much more pronounced. From our assessment of the structural data available in the literature, both trends are captured for Pb-based compositions. The Pb-free FASnI₃ perovskite presents a unique case, as it shows by far the highest thermal expansion rates over the temperatures and compositions explored in Fig. 4, and among the highest recorded for any inorganic extended crystalline solid⁴⁷. Notably, this goes against the general trend of metal–halide binding strength and is probably the result of the relatively stronger stereochemical activity of *ns*² electron pairs in Sn(II) (ref. 20).

What is striking in Fig. 4 is that an extremely strong phase dependence also emerges^{82,85}. For the example of the FAPbI₃ perovskite, Fabini et al.⁸⁵ showed that the value of α_v in the tetragonal phase is roughly double that in the cubic phase (Fig. 4), indicating that devices made with high-symmetry FAPbI₃ crystals are endowed with preferable thermo-mechanical properties¹¹. The higher-symmetry phases of CsPbI₃ and CsPbBr₃ track a similar trend, whereas the MA-based perovskite shows the opposite. Furthermore, the cubic-to-tetragonal phase transition in MA-based perovskites⁸⁶ is accompanied by a large discontinuous volume drop as the cation motion becomes frozen and activates additional dynamic tilts³². Notably, whereas the inorganic double-perovskites present comparatively lower thermal expansion rates, phase changes and discontinuous features are markedly absent over this temperature range. Given that large, abrupt changes in the absorber crystal volume are detrimental to long-term stability, it becomes important to identify such behaviour if it exists within the normal temperature range of solar cell operation.

Mixed-cation perovskite compositions are now widely adopted within highly efficient PSC technologies and represent a special situation. For example, state-of-the-art single-junction PSCs are typically based on MA-free, mixed-cation (FACs)PbI₃ perovskites. Whereas one would expect something analogous to Vegard's law to be in effect for such A-site mixing, the interaction of disparate, ordered cations and their dynamics is not necessarily straightforward. Furthermore, as shown by the thermal expansion data in Fig. 4, comparing different compositions across different phases and temperature ranges is simply inaccurate. Fortunately, Wang et al.⁸⁷ reported a comparable high-symmetry α -phase across the entire FA_{1-x}Cs_xPbI₃ mixed-cation composition space (by creating quantum dots) and reported the thermal expansion characteristics over equivalent temperature ranges. These data are shown in Fig. 5. Considering that the rate of volumetric thermal expansion of α -CsPbI₃ is distinctly larger than α -FAPbI₃ (Fig. 4), the FA_{1-x}Cs_xPbI₃ compositions yield a relatively smooth interpolation between the expansion rates for the pure components. Notably, there is bowing present in the trend with mixed compositions sitting slightly above a linear fit between the two, indicating elevated disorder and/or anharmonicity in the mixed-cation phase.

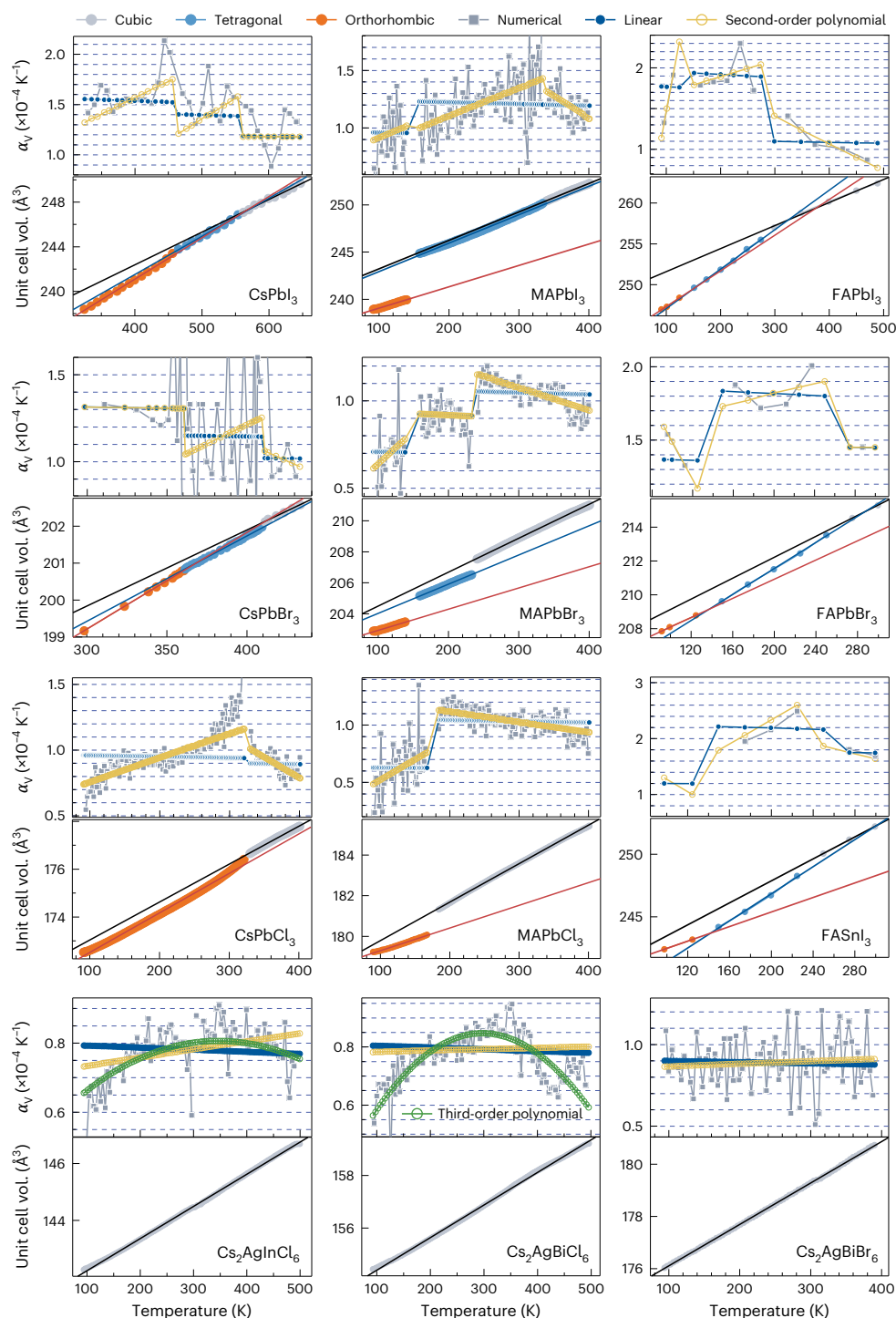


Fig. 4 | Phase-dependent thermal expansion rates. A comparison of the pseudo-cubic unit cell volume for the different perovskite compositions and phases shown in Fig. 3d. The unit cell volumes have been fitted with a linear function for each phase to act as a guide for the eye and to highlight shifts in the rates of

change. The derived volumetric thermal expansion coefficient (α_v) data have been determined numerically (grey filled squares) and using first-order (filled blue circles), second-order (yellow open circles) and third-order (green open circles) polynomial fits (Methods). Data from ref. 80.

Phase-dependent anisotropic thermal expansion

Regulating the direction and distribution of grain orientations in polycrystalline perovskite thin films, that is, the so-called crystal texture⁸⁸, can help to increase device efficiencies; however, the thermo-mechanical consequences of using highly oriented thin films (Fig. 6a) remains an open question. Whereas the underlying origins of texture formation are still relatively ambiguous, the planar nature of the perovskite–substrate interface acts to preferentially select the formation of certain grain orientations over others⁸⁸. Consequently,

the reporting of high-performance PSCs using highly oriented crystal grains is now relatively common⁸⁹.

This means that understanding the anisotropic thermal expansion in thin-film solar cells is vital when attempting to minimize thermally induced stress. From an experimental viewpoint, crystallographic techniques offer a direct measure of anisotropic structures: the evolution of the perovskite unit cell parameters describes its bulk mechanical response with changing temperature (or pressure). Taking MAPbBr₃ (ref. 81) again as an educational example, we show the relative thermal

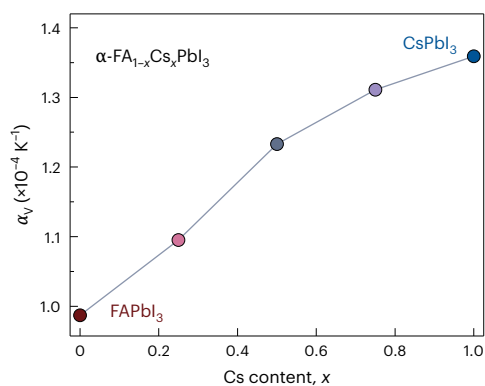


Fig. 5 | Mixing of A-site cations. Volumetric thermal expansion coefficients of $\text{FA}_{1-x}\text{Cs}_x\text{PbI}_3$ perovskite quantum dots with varying degrees of cation mixing. These single coefficient values are from ref. 87, which measured the high-temperature thermal expansion (30–170 °C) for all compositions in a comparable cubic perovskite phase.

expansion/contraction along its principal axes in Fig. 6b (which align to the unit cell axes for the α -, β - and γ -phases). Predictably, crystal symmetry plays a crucial role in determining the changes in the thermal expansivity, with the lower-symmetry phases exhibiting increased anisotropy. In Fig. 6c, the thermal expansivity tensor is represented visually in the form of the expansivity indicatrix⁹⁰; the expansivity is a smoothly varying surface centred at the unit cell origin. From this perspective, it is important to consider the (most likely) orientation of grains in the polycrystalline film, and their phase(s), to accurately determine misfit strains in PSCs during thermal cycling.

Perspectives and outlook

We have taken a comprehensive look at the central role that anharmonic phonons play in shaping the high rates of thermal expansion found in perovskite photovoltaics and related thin-film technologies. Despite their promising energy conversion efficiencies, large thermal strains within PSCs represent an important engineering challenge¹¹. Historically, commercial thin-film devices typically undergo a research-and-development trajectory—increasing the PCE and stability, and upscaling—before making it to the marketplace. PSCs are no exception. As most studies focus on judging whether phonon anharmonicity exists or not, and estimating its relative strength, developing universal criteria for quantifying⁹¹ the system becomes important for generating effective design principles. With this in mind, we provide a consolidated overview of the different avenues that can be leveraged to mitigate thermal strain in PSCs, understand their origins and propose testing protocols for clarifying this inherent degradation pathway.

The perovskite absorber

It is essential to engineer perovskite absorbers that exhibit desirable thermal expansion properties to regulate stress and preserve PSC performance over repeated day/night cycles (Fig. 1c). Notably, simple calculations based on the thermal expansion mismatch (equation (1)) can fail to accurately predict residual stress in solution-processed polycrystalline films, due to dynamic interfacial attachment⁴¹. Thus, it is important to directly measure and establish clear quantifiable boundaries for strain tolerance in the perovskite film¹⁴. For example, the thermal cycling (from –60 to 80 °C) of triple-cation PSCs ($\text{Cs}_{0.05}(\text{FA}_{0.98}\text{MA}_{0.02})_{0.95}\text{Pb}(\text{I}_{0.98}\text{Br}_{0.02})_3$) was reported to exhibit a stable cycling of strain in a narrower range (from –0.06 to 0.38%), that is, the crystal structure was recovered with each cycle and the lattice strain was released. On the other hand, undergoing larger strain evolutions (from –0.13 to 0.57%) introduced non-recoverable changes in the film, that is, plastic deformation or film cracking, which accelerated performance decay¹⁴.

In contrast to other compositions, FA-based perovskites exhibit suppressed anisotropic expansion with changes in temperature, probably due to their bulky size and steric hindrance effects. Notably, the only known region of NTE for FA-containing compositions is in the low-temperature γ -phase of FASnI_3 (Fig. 4). Together, these properties of FA-based perovskites facilitate a relatively smooth build-up and release of thermal strain through each cycle. Angel et al.⁹² suggest that this emerges from the fact that the octahedra are more compressible than the extra-framework cation sites. With the FA cations in a glass-like state, dynamic structural fluctuations are curbed by an inability of the crystal to lower its symmetry with decreasing unit cell volume⁸⁵. Similarly, an effective strategy to alleviate stress in the perovskite crystal involves incorporating bulky molecules, such as β -poly(1,1-difluoroethylene), which possess an ordered dipole structure. Functioning as molecular springs, they dynamically respond to stress, better accommodating mismatch strain in the polycrystalline perovskite film¹⁴.

Thermal expansion is generally reduced by increasing the B–X bonding strength, particularly in the hybrid systems, as it sterically hinders the movement of the organic cations⁹³. What is surprising in Fig. 4, however, is that even for a well-defined ABX_3 perovskite composition, the volumetric thermal expansion coefficient (α_v) varies dramatically across each thermal-phase domain. The value of α_v in the cubic phase of the FAPbI_3 perovskite ($\sim 100 \times 10^{-6} \text{ K}^{-1}$), for example, is about half that in its tetragonal phase ($\sim 200 \times 10^{-6} \text{ K}^{-1}$), indicating a promising phase-engineering strategy for limiting thermal-driven stresses. In practice, Shen et al.¹¹ engineered a higher-symmetry FAPbI_3 perovskite crystal which exhibited retarded performance decay under day/night cycling¹¹.

FACsPbI_3 mixed-cation systems have become increasingly popular in high-performance, MA-free PSCs, as Cs counteracts the limitations of the non-ideal FAPbI_3 tolerance factor near room temperature. Thus, the trend presented in Fig. 5 indicates that, whereas incorporating Cs into FAPbI_3 -based perovskites can enhance phase stability, it will come at the cost of increasing α_v in the absorber. Likewise, on the basis of the thermal expansion rates of FASnI_3 and FAPbI_3 , introducing Sn into a Pb-based perovskite to yield narrow-bandgap absorbers will probably result in elevated thermal expansion rates, and one must consider such changes when designing absorbers using this popular strategy. In general, the mixing of A-, B- and X-sites in state-of-the-art MHP presents a unique engineering challenge when it comes to managing thermal mismatch. This is because the exact absorber compositions vary substantially between different reports, and there are scarcely enough thermo-mechanical data available to develop generalized guidelines. We believe that this is where modern computational tools⁹⁴ can be helpful, by compiling databases of known and predicted MHP compositions and by conducting high-throughput first-principles studies to systematically explore their phonon anharmonicity, local disorder and thermal expansion behaviour across a broad chemical/phase space.

Following changes in the crystal composition or symmetry, it is apparent that shifting phase-transition temperatures outside the operating range of the solar cell will mitigate abrupt changes in the thermo-mechanical properties of the absorber. This is particularly relevant for MA-based materials, which undergo the largest, discontinuous changes in α_v at phase-transition points, due to their large dipole strength (compared with FA and Cs cations) and dynamically ordered interactions with the halide cages. For example, MAPbI_3 experiences transient periods of rapid (non-linear) and anisotropic thermal expansion²⁶ during each thermal cycle, which will induce large and poorly accommodated strain evolutions. It follows that, in general, using higher-symmetry phases¹¹ will inherently limit the impact of anisotropic expansion, grain orientations and NTE within polycrystalline perovskite films.

The extreme anisotropy of the thermal expansion rates in the low-symmetry perovskite phases in Fig. 6 calls for consideration of the

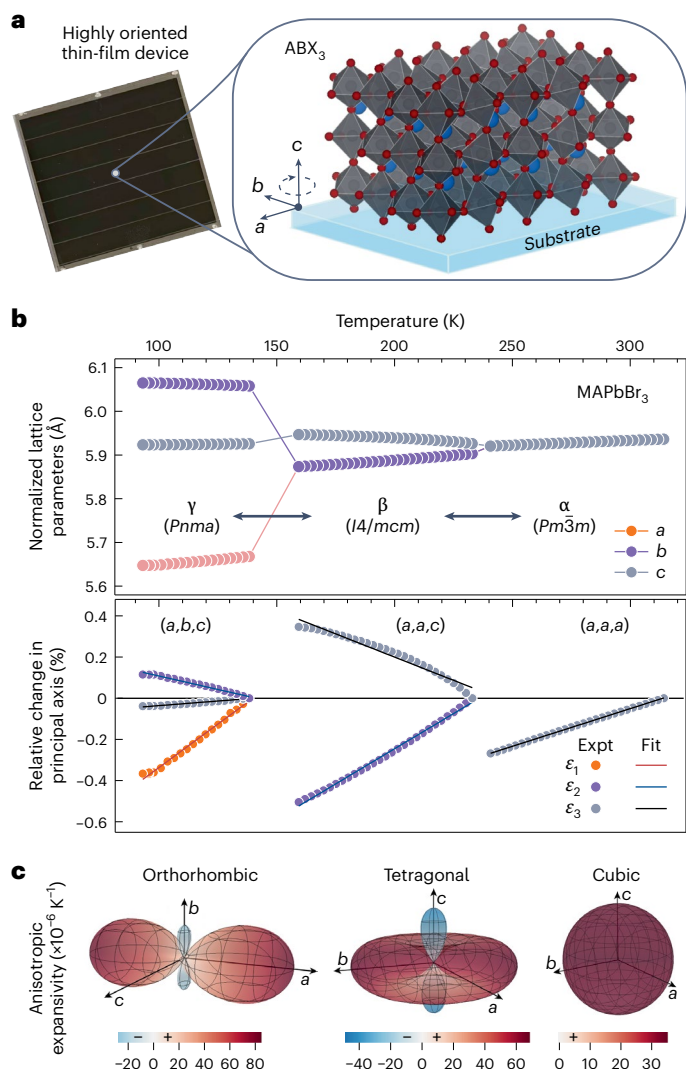


Fig. 6 | Anisotropic expansion in an oriented thin film. **a**, Photograph of a PSC, with the expansion illustrating a highly oriented perovskite absorber with the $\langle 001 \rangle$ crystallographic direction orientated out of plane. **b**, Top: temperature-dependent pseudo-cubic lattice parameters (a, b, c) of MAPbBr₃. Bottom: corresponding relative changes in the three principal axes of the perovskite unit cell (represented as Eulerian strain tensor elements, ϵ_i) over each temperature range of its three perovskite phases. **c**, Visualization of the principal components of the anisotropic thermal expansivity tensor for the different phases of MAPbBr₃. Data in panels **b** and **c** from ref. **81**.

grain orientations within the absorbers. Furthermore, such extreme anisotropy indicates that utilizing deposition techniques to orient the largest thermal expansion axis out of plane will allow it to expand/contract with less resistance, and limit the thermal stress by aligning the less-sensitive expansion axes with the substrate plane, that is, resulting in a smaller mismatch in between layers.

Other device layers

Beyond the absorber, other layers can be engineered to improve durability under dynamic operating conditions. A practical solution involves selecting materials with thermal expansion coefficients that closely match those of the perovskite, or engineering a thermal mismatch operating point that enables a favourable evolution of strain during thermal cycling of the device⁹⁵. In particular, thermal-induced compressive stain (contracted in plane) and tensile strain (expanded in plane) of the perovskite lattice have been shown to have different impacts on solar cell performance. Specifically, it is tensile strain that imposes a marked

reduction in performance⁹⁵. Taking room temperature as the reference point, the α_c values of pure MHPs (Fig. 4) vary from roughly 10×10^{-5} to $20 \times 10^{-5} \text{ K}^{-1}$. This is extremely high compared with the values of typical substrates (for example, glass and indium tin oxide at $-2 \times 10^{-5} \text{ K}^{-1}$, Si at $-1 \times 10^{-5} \text{ K}^{-1}$) and inorganic electron transport layers (for example, SnO₂ at $-1 \times 10^{-5} \text{ K}^{-1}$, TiO₂ at $-2 \times 10^{-5} \text{ K}^{-1}$), and relatively small compared with hole transport layers such as P3HT^{96,97} ($>80 \times 10^{-5} \text{ K}^{-1}$) and poly(3,4-ethylenedioxythiophene):polystyrene sulfonate (PEDOT:PSS) ($>100 \times 10^{-5} \text{ K}^{-1}$). Moreover, with uneven illumination and heat exposure, strong lattice anharmonicity in the MHP absorber drives rapid phonon decay and suppresses thermal transport, causing localized heating^{82,98}. Incorporating thermally dissipating layers can even out temperature gradients and reduce concentrated thermal stress. Together, this indicates that, depending on the device architecture (for example, p-i-n or n-p-i single-junction designs, or tandem devices) and fabrication sequence, considerations can be made to minimize thermal mismatch or ensure that tolerable degrees of compressive strain manifest in the film during operation⁹⁹.

Alternatively, attempts have been made to introduce a relatively thin intermediate layer ($<10 \text{ nm}$) that can bridge the stress mismatch⁹⁵. For instance, introducing a dynamic interface made of a chiral perovskite interlayer⁹⁹ has recently proved effective in strengthening the contact between interfacing layers. This helps to curb delamination between layers, and the use of a chiral interlayer ultimately acts to minimize device fatigue during repeated thermal cycling⁹⁹.

Using a polymer substrate^{95,100} or buffer layer¹⁴ can also counter thermo-mechanical stress and results in only minor, fully recoverable strain variations. Integrating interlayers composed of flexible bridging molecules¹⁰¹ (so-called self-assembled monolayers) between the perovskite and substrate has become a popular choice recently, promoting more stable interfaces and reducing the risk of mechanical failure. Thus, mechanical strengthening of the perovskite–substrate heterointerface enables an even strain profile across the absorber¹⁰² and curbs strain hot spots that lead to delamination. Notably, however, there still remains debate about the actual role of relatively thin and soft contact layers¹⁰⁰; thinner layers are likely to exert negligible force, making them ineffective at changing the strain in perovskite films, especially when a thick, stiff substrate dominates the mechanical response.

The use of non-planar device structures can potentially reduce stress accumulation compared with a normal planar design¹⁰³, yet the specific stress distribution depends on the geometry of the non-planar surface. Specifically, pyramid-textured Si subcells are commonly used in tandem devices to improve light trapping, while also introducing a non-planar interface for the thermal mismatch strain to form, that is, the local thermal mismatch vectors will be redistributed out of plane. Furthermore, ensuring no sharp angles at the rough interface¹⁰⁴ can also alleviate stress concentration issues during the thermal cycling of pyramid-textured Si/PSC designs.

Testing and reporting procedures

It is clearly important that the standardized testing protocols of PSCs are expanded to include the routine reporting of thermal expansion characteristics and device-cycling failure points. Whereas protocols for evaluating performance under thermal cycling do exist¹⁵, they often involve temperature ranges that are too narrow for certain real-world applications and provide limited guidance on factors such as heating and cooling rates. This issue extends to the testing of devices intended to operate under extreme conditions, such as outdoor deployment in relatively cold/hot climates and for applications in space. Notably, recent studies have started to explore the role of thermal cycling down to relatively cold temperatures (-100 K)¹⁰. In general, the same arguments outlined above with respect to phase changes, expansion rates and thermal mismatch still apply at extremely low or high temperatures, and the protocols required to stress test devices that are destined for extreme environments should be set accordingly.

To better replicate broad and rapid temperature changes between day and night, such as those caused by sunlight exposure or shading, accelerated testing conditions with broader thermal ramp rates were recently prescribed by Abate and colleagues⁹; a protocol featuring a ramp rate of 20 °C min⁻¹ with a 2 min equilibrium hold between -60 and +80 °C can offer a more realistic simulation of extreme environmental fluctuations. This approach has proved effective in linking accelerated perovskite degradation to reduced thermal cycling stability¹⁴. Within this context, identifying the failure mechanism(s) involved as a result of the thermal cycling should be reported. This includes explicit descriptions of the device performance metrics beyond the PCE, such as the underlying decline in FF, V_{oc} and short-circuit current density (J_{sc}).

Simple empirical tests do not, however, directly yield insights into the thermo-mechanical properties of the device or reveal how PSCs degrade when subjected to temperature changes. Ultimately, whereas each device architecture will respond differently to thermal fluctuations⁴¹, reporting the thermal and strain tolerances together¹⁴ of thermally cycled PSCs—thus identifying regimes of unrecoverable strain and degradation—is critical for enabling well-informed design principles. Immense scientific value is also added through routine temperature-dependent lattice parameter analysis¹⁰⁵ of mixed perovskite compositions, by directly evaluating their thermal expansion characteristics. With this Review representing the bulk of structural data available in the literature today, such data will help to fill an important knowledge gap and provide a roadmap towards PSCs that combine high efficiency with long-term durability.

It is our hope that the broad range of insights provided here will help to bring clarity to the unusual lattice dynamics and thermo-mechanical properties of PSCs, and provide a clear framework to regulate thermal strain within state-of-the-art device architectures moving forwards.

Methods

Determining thermal expansion rates

For solids that exhibit temperature-independent thermal expansion we expect to observe a near-linear change in the unit cell volume (ΔV) due to variation in the temperature (ΔT):

$$\alpha_v = \frac{1}{V} \frac{\Delta V}{\Delta T}. \quad (2)$$

If α_v is independent of the temperature, this yields the linear equation $V(T) = V_0(1 + \alpha_v T)$, where V_0 is the lattice volume at $T = 0$ K. However, halide perovskites frequently exhibit non-linear rates of volumetric thermal expansion—even for a single phase. In this scenario, a single α_v value is not sufficient to accurately describe the thermal expansion and we must integrate over the experimental temperature range:

$$V(T) = V_0 \exp \left[\int_{T_1}^{T_2} \alpha_v(T) dT \right]. \quad (3)$$

Here, $V(T)$ is a second-order polynomial function that describes the lattice volume as function of the temperature: $V(T) = V_0 + k_1 T + k_2 T^2$, where k_1 and k_2 are the coefficients of the polynomial. Solving equation (3) thus provides the instantaneous coefficient of thermal expansion at a given temperature as

$$\alpha_v(T) = \frac{1}{V} \frac{dV}{dT} = \frac{k_1 + 2k_2 T}{V_0 + T(k_1 + k_2 T)}. \quad (4)$$

An evaluation of the volumetric thermal expansion averages out the anisotropic properties of the crystal expansion, yet similar descriptions to the equations above can be applied to the 1D thermal expansion coefficients (α_i) along each lattice direction separately.

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Competing interests

The author declares no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Julian A. Steele.

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